

Risk Management in Financial Institutions with applied Machine Learning

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Abstract— In today's business world, technological applications are becoming more important in management. Among the most prevalent influencers in business applications are machine learning, artificial intelligence, and other algorithmic applications. They offer a wide range of fixes for issues with business management, risk management in banking included. In the past ten years, risk management has become more important in the financial services industry. Banks used to concentrate on risk assessment, measurement, and reporting. Nevertheless, they are now using machine learning to improve management's efficiency and precision. It determined the areas of risk management that needed attention and investigated several solutions. The need for funding fluctuates depending on the loan provider and is cyclical. Financing must consider the asset's supply and demand in order to guarantee the asset's success. One type of switch-over exercise that entails fast exchanges for cash loans is finance. This study looked into how ML might affect a bank's risk management practices. Essentially, the study showed how machine learning techniques increased the forecast accuracy of risk management models. The techniques therefore performed better than the traditional statistical models. They lessened the negative consequences of sample biases associated with conventional statistical methods.

Keywords— Machine learning, financial institutions, Risk management

I. INTRODUCTION

The field of AI that focuses on teaching computers to automatically change and grow from data or prior experience is called ML [9]. An algorithm that outlines a set of steps to convert input into output can be used to do this [10]. Algorithms are employed in ML to discern between important and unnecessary trends in data. Reliable medical diagnosis, real-time map-based environmental disaster monitoring, and sensory monitoring in industrial processes are a few instances for ML applications. Owing to the financial services they offer, banks are important for economic growth. One could say that their job as intermediaries acts as a stimulant for economic expansion. Any country's financial stability can be measured by the banking sector's long-term, efficient, and effective operations [11].

The rate of a country's economic growth and long-term viability are accelerated by the degree to which banks lend money to the general population for constructive uses. This

relates to lending, which is about taking chances, gaining access to default and interest rate fluctuation risk. It is focused on providing credit facilities to clients, who may be individuals or corporate entities. [12] states that bank credit makes up a significant portion of the labour that commercial banks do, if not the majority of it. Credit facilities are regarded as a significant source of bank returns in addition to making up a sizable amount of the assets of commercial banks. However, a bank's ability to lend money is constrained since it is based on the discrepancy between the total amount of deposits and the portion that the bank retains in liquid cash to satisfy customer demands.

In a socioeconomic system, banks play an important role, and a nation's financial advancement is both a cause and a result of their degree of advancement [17]. The well recognised operations of banks provide proof that alterations in their conduct frequently signify alterations in the habits of their clientele. In addition to collecting savings from people and supporting the financial system through loan activities, banks have played a more significant role in payment services in recent years.

Nevertheless, new technologies have disrupted conventional processes and altered the banks' role in the banking and financial services industry, mostly because of the competition from new operators like Fintechs [14]. With the advent of online banking and the digitization of services, the internet served as a breakdown function in the early stages. However, in recent years, the financial industry and banking activity have benefited greatly from the application of AI and ML [15][16]. Even while banks are accustomed to using technology, the shift to AI and ML implies a really innovative strategy for banking management across the board, from new service offerings to risk management, from trading platforms to customer relationships. There is a vast range of uses and applications, and there is always room for challenge in an environment that gets more complicated.

During the global financial crisis, risk management at banks has become more important, with a constant focus on how risks are recognised, quantified, stated, and handled. Innovations in risk management and banking, along with current and future issues, have been the focus of both industry and academia [18]. Simultaneously, ML is becoming more common in business applications; numerous

solutions are in use right now, and numerous others are being researched.

Big data contributed significantly to the expansion of several firms, and this was made feasible by advancements in the IT industry. Big data is the accumulation of information that a computer can acquire, handle, store, and analyse quickly. The productivity of data processing, exchange, and storage has increased because to big data [19]. One can view all of an company's financial, business, and associated data. It provides vital information for the early detection of internet credit problems.

When big data analytics techniques are used in credit risk management, early warning and risk calculation become more accurate and objective. Credit risk is analysed using the multiple and LR discriminant analysis techniques. Its shortcomings include its restricted dynamic early warning capabilities, excessive reliance on historical data, and requirement for a substantial amount of real data as the premise [13].

Due to its function in bank operations, such as taking deposits and making loans while simultaneously monitoring the potential for some customers to forfeit all or part of their deposits, bank credit is also significant. Stated differently, the money to be approved for credit is the amount that depositors are unlikely to take out during the loan payment period. In order to mitigate any risk associated with the potential for insolvency, banks thus offer loans against monies placed with them at an interest rate higher than that which is offered to consumers for their deposits. It is obvious that a bank's decision to approve credit is significant. Since a poor credit choice will ultimately lead to loans that are nonperforming. Because of this, banks need to be informed of the possible consequences that could arise from a client's failure to make payments. Nevertheless, there are numerous questions about whether financial institutions (FIs) in emerging nations are using scientific methods to anticipate credit risks, even with advances in technology and science [20].

II. LITERATURE REVIEW

ML is becoming more and more prevalent in corporate applications; some solutions have already been put into practice, and numerous others are being investigated. The advancements in risk management and banking, as well as the difficulties that lie ahead, have been the subject of extensive research in industry. In order to recognise fields or issues related to risk management that have not been sufficiently explored and may benefit from more study, this paper reviews the literature in order to assess and analyse ML methods that have been studied in the context of banking risk management [1]. The evaluation demonstrates that while the use of ML in the management of banking risks, including credit, market, operational, and liquidity risk, has been investigated, it doesn't seem to be in line with the industry's current emphasis on risk management and ML. There are still a lot of fields in bank risk management that may gain a lot from research on applying ML to particular issues.

We examine how risk management is changing as a result of AI and ML technologies. First, a non-technical summary of the primary AI and ML methods applied to risk management is provided. Next, an overview of how these strategies have been applied to the risk management domains of credit risk, and compliance is given, supported by empirical data and actual practice. We wrap up with some

observations about the field's present constraints and predictions for its short- to medium-term development [2]. Overall, we paint a positive picture of the application of AI and ML to risk management, but we also point out certain real-world drawbacks, such as the need for appropriate data management practices, transparency, and the shortage of requisite skill sets inside firms.

The financial industry, as well as the economy at large, is increasingly impacted by AI and ML. Following the global financial crisis, the effects of AI and ML on banking risk management have drawn a lot of attention. The potential of AI and ML to further enhance banking risk management is the main emphasis of the research. The goal of the paper is to investigate the likelihood of effective implementation while accounting for various obstacles and issues as well as remedies [3]. The COVID-19 pandemic and other current global economic and financial difficulties may be mitigated in part by the use of AI and ML. This research primarily focuses on credit risk management, but it also analyses the applications of machine learning and artificial intelligence in other risk management domains. It is determined that controlled and well-prepared further application of AI, ML, DL and BDA can have further positive consequences, particularly in the subsequent risk management fields: credit, market, liquidity, operational risk, and other associated fields.

Financial risk management is crucial for the majority of businesses since it reduces losses and increases profits. With the task's strong dependence on information-driven decision making, ML offers the creation of new methods and instruments. In recent years, ML approaches have gained increasing popularity for a range of risk management applications. However, investigators often struggle to make sense of the vast and varied subject knowledge as well as the quickly evolving literature. This paper bridges this gap by providing a comprehensive overview of the rapidly growing corpus of literature on ML research for financial risk management [4]. The research offers four unique contributions: First, we provide taxonomy of financial risk management positions along with the ML methods that go along with them. Second, we present notable works from the previous ten years. Thirdly, we enumerate the primary challenges that scientists working in this area need to solve. Finally, we highlight recent advances and promising directions for further study.

AI in the banking sector has the potential to resolve issues with the current market structure, such as fraud, credit crises, and financial risk management. Additionally, AI could replace the repetitive and routine operations carried out by people in the early stages of financial activities, hence lowering the rates of errors committed by humans and improving the overall transparency of each trading [5]. For credit rating and loan choice problems, this work uses both supervised and unsupervised learning techniques, like SVM, logistic regression, RF, and K-means. Based on the experimental outcomes, we can infer that among these supervised learning models, the SVM algorithm performs the best, with an accuracy of up to 78.4%. Furthermore, the three K-means clustering results are preferable and offer a useful solution for handling SME credit rating. This study provides an AI-based approach to credit scoring and loan decisions, which can assist the banking sector in resolving financial problems.

Cryptocurrency is one of the most popular financial states in the world, and it comes with a lot of risks that impact risk auditors' intrinsic evaluation. The financial sector has always been at risk from the growth of cryptocurrencies due to possible money laundering. Banks and bank secrecy serve as risk specialists, bank managers, and compliance officers in the setting of financial support organisations like anti-money laundering. They are entrusted with looking into connected cryptocurrency transactions and users who conceal illegal funds [6]. This study applies unsupervised ML and Risk Parity to the cryptocurrency system. The professional accounting procedure with relation to the inherent risk associated with cryptocurrencies, identifying the hazards associated with cryptocurrencies, which include unauthorised access to private keys and a high probability of occurrence. When compared to those with less experience, those with professional cryptocurrency transaction experience carry a lesser risk. When it comes to delivering the adjusted risk tail and producing better risk management outcomes, the Hierarchical Risk Parity method performs better. The results section demonstrates how resilient the suggested model is to covariance window estimates and different rebalanced intervals.

III. RESEARCH METHODOLOGY

The demand for financing is cyclical, varying from one loan provider to another. In order to ensure the asset's success, financing needs to take into account both the asset's supply and demand. Finance is one kind of change-over exercise that involves quick swaps for cash loans. From the standpoint of asset demanders, venture financing is a financial movement that uses several budgeting techniques, including trading defences and allocating all available resources, and completes the advancement of operational property exercises. Differential enhancement tactics will be utilised to different tiers of initiatives based on their development needs. The first step in putting the funding model into practice is to promote the useful and eye-catching events they host.

Banks are hesitant to extend more loans, and a company's early efforts cannot support high personnel expenses. The main force behind venture finance is the accumulation of specialised capital. This study examined and pre-processed the large-scale data before training and evaluating three models—KNN, LR, and XG boost—for their ability to forecast loan defaults and their likelihood. The precision, recall, F1, and ROCAUC measures are used to assess the models' ability to predict class labels.

Big data is defined as complex data types and large amounts of data. Hadoop is a well-liked system for handling massive data batches. Regression makes use of well-known statistical methods, such as the most basic kind of linear regression. However, many issues are outside the scope of straightforward linear projections of historical data. For instance, predicting product sales volumes, stock prices, and failure rates can be difficult due to intricate relationships between many predictor characteristics. It may be necessary to use more complex algorithms in order to forecast future data. The model types used in regression and classification are frequently the same.

LR is a widely used statistical method for classifying and connecting situations. The LR model works well for credit rating problems because it uses a different distribution

function for estimation than other statistical techniques. Various techniques for building a binary logistic regression model have been offered in order to improve the model's accuracy and versatility.

Regression predictive modelling can make use of XG boost, an implementation of sharp gradient boosting. Repeated k-fold cross-validation is a best practice approach for testing an XG boost regression model. The most common algorithm is ML, regardless of whether the problem at hand is a regression or classification one. It is commonly recognised that it performs better than any other ML method. An implementation of the gradient-boosting decision tree approach is included in the XG boost package.

It is a well-liked and effective open-source gradient-boosted tree implementation. Gradient boosting is a supervised learning method that makes somewhat accurate predictions about a target variable by aggregating the predictions of multiple smaller, weaker models.

Precision and mistake rates are important measures of categorization systems in financial risk prediction. The performance score is calculated using measurements of the receiver operating characteristic curve area under the curve, general precision, true positive rate, and true negative rate.

IV. RESULTS

TABLE I. K-NEAREST NEIGHBORS (KNN):

Metric	Score
Precision	.85
Recall	.78
F1 Score	.81
ROC AUC	.88
Overall Accuracy	.82
True Positive Rate	.75
True Negative Rate	.89

TABLE II. LOGISTIC REGRESSION (LR):

Metric	Score
Precision	.88
Recall	.82
F1 Score	.85
ROC AUC	.90
Overall Accuracy	.84
True Positive Rate	.80
True Negative Rate	.88

TABLE III. XGBOOST:

Metric	Score
Precision	.90
Recall	.85
F1 Score	.87

ROC AUC	.92
Overall Accuracy	.86
True Positive Rate	.83
True Negative Rate	.89

TABLE IV. OVERALL MODEL PERFORMANCE

Metric	KNN	LR	XGBoost
Precision (Average)	0.78	0.84	0.87
Recall (Average)	0.82	0.79	0.85
F1 Score (Average)	0.80	0.81	0.86
ROC AUC (Average)	0.85	0.87	0.90

V. DISCUSSION

It has been demonstrated that ML techniques outperform conventional statistical techniques in terms of both predictive and classification accuracy. It is believed that the support vector machine is a well-researched and effective ML technique. A lot of empirical research uses data from observations. Because of sample design or sampled unit behaviour, selection methods may produce non-random samples. In the former, missing data are common, while in the latter, sample units self-select. Bias in sample selection results from this [7]. It is said to be "choice based" when the sample data under study contains a proportionate representation of some dependent variable results that differs from its proportionate representation in the population from which it is taken. The computation is biased as a result of this "choice based" sampling. ML may be vulnerable to the same issues and biases that plague conventional statistical techniques because a large portion of its modelling is based on data that is already available. It would be helpful to assess and comprehend how issues with conventional statistical research approaches are handled by ML strategies as they are contrasted with ML procedures.

Nor does it seem that the use of ML for market risk management has received enough attention. Numerous studies have examined market risk or volatility from the standpoint of portfolio or investment risk management. Nonetheless, the papers seem narrow when seen through the lens of bank risk management. It is worthwhile to conduct more study on this, particularly from the perspective of stress testing or tail end risk capture. A few use cases have been studied for liquidity risk, which has drawn significant attention from regulators since the financial crisis.

Research in the field of operational risk have primarily concentrated on the detection of fraudulent and suspicious transactions—issues that are generally handled by classification algorithms. Another term for neural networks is "a very popular and renowned method in credit card fraud detection."

Aside from applications connected to money laundering, a large portion of the other fields of non-financial risk management, country risk management, and compliance risk management haven't been sufficiently investigated. Research publications on conduct risk appear to be limited, despite the fact that ML is suggested as a way to handle conduct risk. Conduct risk has emerged as a major concern and been given top priority by regulators and banks due to the wave of conduct issues in Europe, the US, and Asia Pacific. Studies on a number of common risk management methods,

including risk assessments, risk monitoring, and risk reporting, would also seem to be highly beneficial.

Investigators employ a variety of dataset sources for their work; some use databases given by financial services providers (like Moody's) or commercial banks, while others use data that is readily available to the public for their study. Issues brought to light include the possibility of extremely skewed data (such as low default rates), inaccurate information, and inconsistent data integrity (such as incorrect or erroneous labelling) [8]. Greater access to real-world data sets would stimulate additional studies assessing the various issues risk management services face.

VI. CHALLENGES

Prior achieving the full potential of AI and ML approaches for risk management, there exist certain noteworthy practical challenges that require attention. The most crucial of them is that appropriate data are readily available. While ML packages for R and Python can easily read any kind of data, from images to natural language processing to calling data from SQL databases, ML solutions have not been proposed as quickly as firms have been able to appropriately organise the internal data they have access to. Data is often stored in distinct silos across departments, maybe on several platforms, and possibly with internal political and legal constraints limiting data exchange [21].

Crucial information may not even be preserved on file as official data, but rather retained as unofficial company knowledge. The availability of qualified personnel to put these new methods into practice is another problem. The readiness and capacity of staff to comprehend and operate with these novel technologies was identified as the top concern among the top 1000 US corporations surveyed about the use of AI. Transparency is the last significant challenge that AI-driven solutions must handle. The models for the DL approach of ML operate on hidden layers that exist between the input and output data. Such a black box system can lead to problems with regulatory compliance, when it comes to proving model validity, and is not conducive to efficient risk oversight. A more theoretical concern associated with this is that trading optimal models used by several organisations may converge to comparable models, resulting in systemic risk.

Setting these concerns aside for the time being, it is worthwhile to look at AI and ML potential applications in risk management from a more optimistic standpoint. The obvious implication is that risk management will become far less expensive and time-consuming. For example, 8,000 of the 137,000 employees of BBVA, the second-biggest bank in Spain, are employed in compliance. They are making significant investments in AI technologies to lower this foundation of compliance costs. Businesses who invest more in this field will inevitably see cost savings from AI and ML capacity to automate repetitive operations and to organise, retrieve, and cluster non-conventional data, such as papers [22].

Additionally, AI will provide more precise real-time data on all kinds of risks the company is taking. Real-time guidance will become more and more prevalent as data organisations focus more on using AI. Predictive notice of hazards is the next stage after real-time understanding of risks being incurred. The ability to precisely identify business hazards in advance, whether they be credit, operational, or

market concerns, is, in a sense, the holy grail of AI-driven risk management systems. This capability is provided by ML approaches, which conventional statistical techniques could never hope to match.

A truly AI risk management system, which would automatically step in to prevent unnecessary risks, quickly unwind dangerous exposures, and dynamically modify the firm's risk appetite based on the system's estimation of the broader risk environment, would not be technologically impeded, to think even further ahead. Even still, those risks will still need to be handled because of that, which will keep risk management specialists happily engaged for the foreseeable future (despite a rapidly changing environment).

VII. FUTURE SCOPE

In the upcoming years, AI is expected to continue revolutionising the finance sector, as per many experts. AI is utilised in many intricate ways to analyse data, spot trends and perspectives, streamline tasks, and generate a wide range of suggestions. AI may develop to the point where it can execute complex trading approaches and generate incredibly precise market predictions in the trading and finance industries. The companies might be able to maximise both profits and investments as a result. Nevertheless, as AI becomes more involved in financial decision-making, proper oversight will be essential.

Through data analysis, AI will assist banks in getting a deeper understanding of its clientele, enabling more individualised services. While financial planning and customer service are currently handled by chatbots and robo-advisors, these technologies will soon advance to resemble real beings. Significant developments are also to be anticipated, such as the incorporation of AI-powered solutions into bank digital wallets. Complex AI systems will automate numerous human operations, such as fraud detection and loan application assessment. Human monitoring and governance, nevertheless, will always be essential.

AI is also transforming financial risk assessment and regulation. ML can be used to examine alternative data in order to find events or risks that humans would miss. Therefore, AI may aid regulators in their oversight; but, accountability will be contingent upon the rationale and transparency of the AI systems. However, since hackers are already utilising AI to take advantage of weaknesses, the finance sector needs to be extremely watchful. Ultimately, experts stress that although AI greatly improves efficiency, intelligence, and creativity, human interaction in finance is essential. We need hybrid intelligence systems, which integrate AI with human knowledge, morality, and feelings. This human-AI partnership is where finance is headed.

VIII. CONCLUSION

Finance and banking will be significantly impacted by ML. Risk management is one area where financial algorithms and ML may have an even bigger influence. This paper investigated the potential effects of ML on a bank's risk management procedures. In essence, the study demonstrated how ML methods improved risk management models' forecast accuracy. As a result, the methods outperformed the conventional statistical models. They reduced the detrimental effects of sample biases linked to

traditional statistical techniques. ML approaches, for example, are capable of evaluating massive amounts of data without limitations, like presumptions about the data's distribution. They also possessed the ability to get profound understanding of the hazards, categories, and analytical capacities. For this reason, banks are looking for ML approaches to improve their risk management skills. The methods' higher promise for revolutionising risk management and providing precise risk models stems from their ability to find intricate, non-linear patterns in huge datasets.

These days, AI and ML are hot issues in many corporate domains, particularly the financial services industry. Since FIs must handle and mine ever-increasing volumes of unstructured data and regulatory reporting data for compliance or to efficiently engage with other FIs and FinTechs, they are searching for more potent analytical techniques. It appears that ML and AI are having an influence on each component of the financial industry business model. These technologies have the potential to enhance risk management, client preferences, fraud and carry out breach detection, streamline client assistance, and enable automated identity verification when combined with biometrics.

This paper provided an overview of ML and covered a number of financial institution-specific applications, including credit risk modelling, money laundering and credit card fraud detection, and FI-specific carry out breach surveillance. Its discussions were based on conversations with IIF members and technology vendors. Regarding the application of ML in the financial industry, two hazy conclusions are emerging; this is because the field is still in its early stages of development and many FIs are still experimenting with ML in certain capacities.

Initially, ML encompasses a variety of statistical learning instruments that can typically analyse substantial volumes of data, providing a great degree of granularity and depth of analysis, primarily for predictive reasons. The degree of detail with which data may be examined and results predicted is increased by certain approaches' capacity to infer non-linear correlations and do data analysis without making preconceptions about the form or shape of the relationship among variables (i.e., non-parametric). It is possible to explore data without a dependent variable using unsupervised methodologies. Combining models and running algorithms thousands of times on training data increases the predictive potential of the models while preventing overfitting and preserving analytical granularity. FIs can now obtain greater insights into business activities including lending, risk management, customer interaction, and payments thanks to these enhanced, frequently automated analytical capabilities. ML can identify deeper, more intricate patterns and relationships in the ever-increasing amount of data generated by these processes, such as in the analysis of transactions or credit risk, or it can connect disparate datasets to derive more precise overall conclusions, such as in the monitoring of conduct breaches.

Second, there is a great deal of context-dependency in the implementation of ML techniques in the finance industry. Availability of large, high-quality data in FIs for training or analysis is not always guaranteed. More crucially, there may be a trade-off between increasing model complexity and a lack of explanatory insight and the predictive strength and granularity of analysis of several

methodologies. This is a problem especially when analytics are used in a regulatory setting, and a compliance team or supervisor will want to examine and comprehend the model that was used.

Fortunately, there are less complex ML techniques that combine simplicity and non-linear analysis. In fact, providers of ML analytics for the financial industry often seek to enhance the auditability of their offerings by using factor models or combining the depth of insight provided by ML with the simplicity of models. It appears that every problem has an algorithm.

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